WFLOW-04: Advanced Notification Routing

> **Series:** WFLOW — Workflows and Alert Notifications | **Notebook:** 4 of 10 | **Created:** January 2026 | **Last Updated:** 11/11/2026

Intelligent Alert Routing

Not all alerts should go to everyone. This notebook covers conditional routing based on severity, team ownership, time of day, and escalation patterns.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription

Requirement	Details
Permissions	`automation:workflows:write` ; the workflow actor also needs `storage:events:read` for the Problem trigger
Prior Knowledge	**WFLOW-01** through **WFLOW-03**
Connections	Slack and/or Teams connections configured

Sprint 1.337 (April 2026): Smartscape Ownership for Routing

SaaS 1.337 made ownership available on Smartscape nodes, for use from **Workflows actions**: *"Ownership information is now available in Smartscape and for Smartscape nodes. This lets you use Workflows actions (such as get_owners) to send automated and targeted notifications to your teams"* (What's new in SaaS 1.337 (DT docs)). It is not a DQL field. An earlier revision of this notebook read `getNodeField(affected_entity_ids[0], "ownership.team")` ; that returns null on every problem, because no Smartscape node model has an `ownership.*` field and `getNodeField` does not resolve a classic entity ID.

Two documented routes:

- **Route on ownership with the Ownership app's *Get owners* action.** Add a *Get owners* task after the Problem trigger. Its *Entity ids* input takes *"A Jinja expression, for example,* `{{ event()["dt.entity.host"] }}` *"*, but that field is rarely set on a problem (33 of 4,244 problems over 7 days on the validation tenant, 10/06/2026). Pass the problem's entities instead, `{{ event()["affected_entity_ids"] | join(",") }}` ; the input takes several: *"Use a comma or semicolon to separate multiple IDs."* `affected_entity_ids` still works but is deprecated: *"affected_entity_ids in favor of smartscape.affected_entities"* (Upgrade from Classic problem notification to simple workflows (DT docs)). The action's documented example is a classic entity-ID field, and the page does not say whether it resolves Smartscape node IDs, so check the output of one run before passing `smartscape.affected_entities` IDs instead. Note also that the classic field names only `ENVIRONMENT-0000000000000001` on problems whose entity exists only in Smartscape (1,479 AWS EC2 problems over 7 days on the validation tenant, 11/11/2026), which no ownership lookup can resolve to a team. The output carries `slackChannels` , `msTeams` , `email` and `jira` , and each is a list per team (*"A list of Slack contact details per team"*), not a single channel (Actions for Ownership (DT docs)). Loop the Slack task over `result("get_owners").slackChannels` with the **Loop task** option and take the

channel from each item; run the workflow once and read an item's shape in the Results tab before wiring it. That replaces hard-coded team names in the workflow.

- **Or route on `primary_tags.team`** in the trigger's custom filter or a task condition. *"Dynatrace enriches all derived signals (service metrics, Davis events, and problems) with the same tags."* (Primary Grail fields and tags (DT docs)). The matcher is `primary_tags.team == "payments"`. Verify your entities carry the tag first: on the validation tenant `primary_tags.team` was null on every problem (09/24/2026).

FAQ-21 §5 covers ownership coverage and the Problem fields mapping in depth.

Dynatrace API 1.337: `metadata` removed from the Environment API v1 events endpoints

The Dynatrace API 1.337 changelog lists, under **Environment API v1**, `GET /events` and `GET /events/{eventId}` with *"Changed EventQueryResult schema (application/json) Changed property events Removed properties: metadata"*. This affects only an HTTP task that calls the v1 endpoints `/api/v1/events` or `/api/v1/events/{eventId}` and parses `metadata`; drop that reference. Workflows started by a Problem or Davis event trigger read Grail records, which never had that property, so they are unaffected.

> **Sources:** Dynatrace API changelog 1.337 (DT docs).

---

1. Routing Strategies

> ⚠️ **Migrating from alerting profiles? Duration suppression maps onto the trigger's Minimum duration option.** An alerting profile can delay notification until a problem has been open longer than *N* minutes (`delayInMinutes`) — a common way to suppress transient blips. The problem trigger's **Minimum duration** option (renamed from **Delay** in 08/2026) postpones *"the trigger until the problem has been open for at least the configured duration"* — 5, 10, 15, 30, 60, 120, 240, 1440, or 10080 minutes, evaluated on `dt.duration_marker` (Problem and event triggers (DT docs)). Since its 09/07/2026 rewrite the alert-notification upgrade guide (DT docs) agrees: *"The delay, update, and severity capabilities described in this guide exist*

> *only on the workflow trigger."* Earlier versions of that guide said there was no alternative; that line is gone.
>
> The values are fixed, so a delay that is not one of those nine has to be rounded — up suppresses a little more, down pages a little sooner. Treat a long delay as evidence the alert was the wrong *shape* rather than merely delayed: a profile suppressing 30 minutes of a firing condition is usually describing a burn-rate concern, which belongs in an SLO burn-rate alert (**SLO-04**) or a Davis anomaly detector (**AIOPS-02**). Inventory `delayInMinutes` across every profile and choose a Minimum duration for each; no special migration wave is needed. See **MZ2POL-09** §6.1.

Why Route Alerts?

Problem	Impact	Solution
Alert fatigue	Teams ignore alerts	Route only relevant alerts
Slow response	Wrong team notified	Route to owners
Off-hours noise	Sleep disruption	Time-based routing
Missed escalation	Unacknowledged alerts	Auto-escalation

Routing Dimensions

Dimension	Route Based On	Example
Severity	`event.severity` (1–5)	Critical (1) → PagerDuty, Minor (3) → Slack
Team	`owner` / `dt.owner` tags, or Smartscape ownership	`owner:checkout` → #checkout-alerts
Service	Service name or entity ID	Payment service → payments team
Time	Hour of day, day of week	Weekends → on-call only
Environment	prod/staging/dev tags	Prod → immediate, Dev → daily digest

2. Conditional Expressions

A task condition decides whether a task runs. There is no workflow-level list of named conditions: each task carries **one** condition, written under the task itself. *"You can express task conditions based on the final state of the predecessor task and as a custom condition to implement any custom logic."*

Part	YAML key	What it holds
State condition	`states`	One entry per predecessor task. The editor offers *success or skipped* (the default), *success*, *error or cancelled*, *error* and *any*; exported YAML uses `OK`, `SUCCESS`, `NOK`, `ERROR` and `ANY`, and Dynatrace's samples use `OK`
Custom condition	`custom`	**One** Jinja expression; the task runs only when it is true. Combine checks with `and` / `or` / `not` inside it
Else	`else`	`STOP` or `SKIP`. The default is *stop here*: *"if you want to skip this step and continue the workflow, you can change the else action from stop here (default) to skip"*

A task that follows the trigger directly has no predecessor: *"It's not possible to add a Run this task if and Else condition to the first task."* It can still carry the custom condition: *"Nevertheless, a custom condition can be configured to evaluate the properties of an event trigger context."* Dynatrace's own multi-channel sample nonetheless sets `else: SKIP` on its parallel first tasks in YAML, and the examples in this notebook follow it. In a routing workflow, set Else to **Skip** on every branch that may legitimately not run.

Connection inputs take a connection **ID** (Slack: *"Connection ID."*). In YAML, resolve a connection by its name with `connection()`: *"Get a single connection by schema ID and connection name."* The connection picker in the editor fills the ID for you.

A Condition on a Task

```
tasks:
  pagerduty_alert:
    name: pagerduty_alert
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      # One expression: AND, OR and NOT go inside it
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 and
"env:prod" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"
      # ... event inputs as in §3
```

Negating a Condition

```
tasks:
  slack_only_for_non_critical:
    name: slack_only_for_non_critical
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ not ((event().get("event.severity") | int(5)) &lt;= 1) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-production"
      message: "{{ event()['event.name'] }}"
```

Conditioning on a Predecessor's State

```
tasks:
  notify_on_failure:
    name: notify_on_failure
    action: dynatrace.slack:slack-send-message
    predecessors:
      - create_incident
    conditions:
      states:
        create_incident: ERROR    # run only if create_incident failed
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-integration"
      message: "Incident creation failed for {{ event()['display_id'] }}"
```

Expressions

```
# AND within one expression
{{ (event().get("event.severity") | int(5)) &lt;= 1 and "env:prod" in
(event().get("entity_tags") or []) }}

# Critical or Major (1 or 2)
{{ (event().get("event.severity") | int(5)) &lt;= 2 }}

# Check if field exists (root_cause.smartscape_entity replaces the deprecated
root_cause_entity_id)
{{ event().get("root_cause.smartscape_entity") is not none }}
```

Read any field a problem may lack with `.get()` and a default. The reference documents that reading a missing attribute fails with an *"Undefined variables"* message (shown there for `input()` and `result()`), so the task errors rather than evaluating false. Problem records carry tags in `entity_tags`, an array of `key:value` strings, so `in` tests for an exact element. The field is deprecated (*"`entity_tags` still exists but is deprecated (Dynatrace Classic)"*): prefer the trigger's tag filter (§4) or `primary_tags.*`. An earlier `tags` string field stopped appearing on problem records on 10/08/2026 on the validation tenant, so a condition that reads `tags` is always false and its task skips.

> **Sources:**
>
> - Build workflows (DT docs)
> - Jinja expressions for Workflows (DT docs)
> - Upgrade from Classic problem notification to simple workflows (DT docs) — maps `{Tags}` to *"primary_tags.* entity_tags still exists but is deprecated (Dynatrace Classic)"*
> - **Dictionary:** model `davis.problem` (`dt.davis.problems`) lists `entity_tags` and no `tags` field, read 10/15/2026. Tenant: `countIf(isNotNull(tags))` 0 of 4,282 problems over 7 days, `entity_tags` (array) on 2,076.
> - Slack Connector actions (DT docs)
> - threat-detection-notification-sender.yaml (Dynatrace GitHub) — parallel first tasks with `states: {}` , a `custom` condition and `else: SKIP`

3. Routing by Severity

Severity-Based Routing Pattern

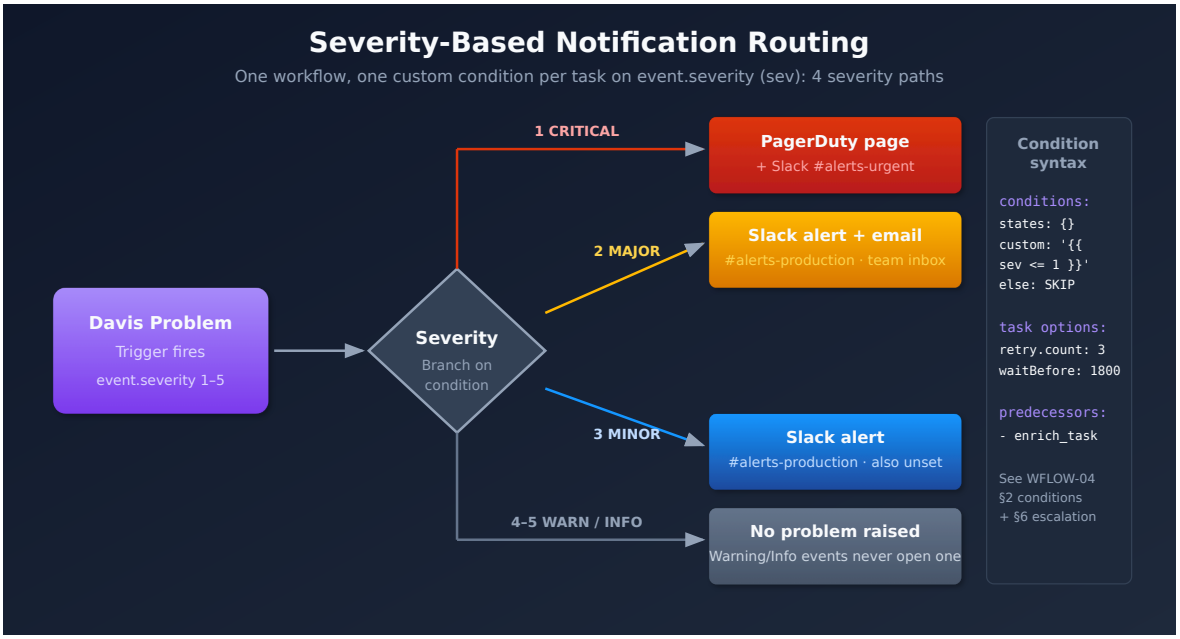
The problem record carries severity as `event.severity` : *"Severity is expressed as a numeric value from 1 (the most critical state) to 5 (purely informational event)."* The levels are named 1 Critical, 2 Major, 3 Minor, 4 Warning and 5 Informational. There is no `CRITICAL` / `HIGH` / `MEDIUM` / `LOW` string on the record, so conditions comparing against those strings are false for every problem.

Problems carry only the first three levels. *"The problem inherits the highest (most critical) severity across all grouped events." "Warning and Informational events appear in the event list and can be grouped into a problem by problem correlation, but they don't raise problems on their own."* On the validation tenant, 3,755 of 4,244 problems in the 7 days to 10/06/2026 carried `3` and the other 489 carried `1` ; none carried `2` , `4` or `5` . Route on the tiers below:

`event.severity`	**Actions**
1 (Critical)	PagerDuty + Slack #alerts-urgent
2 (Major)	Slack #alerts-production + email to the team
3 (Minor), or unset	Slack #alerts-production

To drop low tiers before any condition runs, set the Problem trigger's **Severity** filter: *"Selecting a severity level triggers the workflow for that level and all higher (more critical) levels."*

> **Breaking — SaaS 1.348 (rollout start 10/12/2026, staged tenant rollout): severity is no longer defaulted.** *"Davis events and problems no longer default* `event.severity` *to* `3`*."* A severity filter that was matching the default stops matching once 1.348 reaches your tenant: *"Workflows with a Davis event/problem trigger that filter on* `event.severity=3` *expecting it to be defaulted, might need to be changed to filter for* `Any` *severity to keep being triggered."* Before relying on severity tiers, check which event sources actually **set** a severity. For sources that do not, route on `event.category`, ownership or tags instead, or restore a default with a Davis event OpenPipeline processor. Until 1.348 reaches your tenant, the defaulting behavior still applies and the conditions below keep working. The `int(5)` in them treats a missing severity as the least severe tier, so an unset severity lands with Minor and never pages. **FAQ-21 §6** carries the same note.
>
> **Sources:** Standardized event severity (DT docs), What's new in Dynatrace SaaS 1.348 (DT docs) — GA, rollout start 10/12/2026; re-read 10/23/2026. **Dictionary:** `event.severity` (`stable` , type `long`), read 11/11/2026.



Workflow Configuration

```
tasks:
  # 1 Critical: page on-call + Slack urgent
  pagerduty_critical:
    name: pagerduty_critical
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"   # an Events connection (routing key)
      eventAction: trigger
      severity: critical
      summary: "{{ event()['event.name'] }}"
      source: dynatrace
      dedupKey: "dynatrace-{{ event()['display_id'] }}"

  slack_urgent:
    name: slack_urgent
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-urgent"
      message: ":rotating_light: *CRITICAL* {{ event()['event.name'] }}"

  # 2 Major: Slack alerts (add an email task with the same condition)
  slack_major:
    name: slack_major
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) == 2 }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
```

```
      channel: "#alerts-production"
      message: ":warning: *MAJOR* {{ event()['event.name'] }}"

  # 3 Minor, or unset: Slack only
  slack_minor:
    name: slack_minor
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &gt;= 3 }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-production"
      message: ":information_source: *{{ event()['event.category'] }}* {{
event()['event.name'] }}"
```

Severity levels. `event.severity` is an integer from 1 to 5, with the names Dynatrace uses in the Problem trigger and the problems list:

Integer	Level
1	Critical
2	Major
3	Minor
4	Warning
5	Informational

The conditions in this section key on this numeric field. Earlier revisions compared against `"CRITICAL"` / `"HIGH"` / `"MEDIUM"` / `"LOW"`; the problem record carries no such strings, so those conditions were false for every problem. *"Grail stores severity as a numeric value."* The conditions still convert with `| int`, because the value in the `event()` payload may arrive as a string; Dynatrace's own PagerDuty template likewise normalizes it with `| string | trim` before mapping. The field is `stable` in the semantic dictionary (read 11/11/2026). In DQL (problem feed, reporting) it is queried directly as `event.severity`; see **AIOPS-03 §5** for the severity rollup pattern.

> **Sources:** Standardized event severity (DT docs), Send problem as event to PagerDuty template (DT docs). **Dictionary:** `event.severity` (`stable` , type `long`), read 11/11/2026.

4. Routing by Team/Service

Route by Ownership Tags (recommended)

Dynatrace's ownership keys are built in: *"By default, Dynatrace provides two keys: owner and dt.owner ."* Tag entities with them, then give each team a workflow whose Problem trigger filters on its tag with the trigger's *"Affected entities : Filter by entity tags"* option. The routing decision is then made before any task runs, and no condition is needed. **FAQ-21 §5** walks through this pattern.

Tag Conditions in One Workflow (fallback)

When one workflow must fan out to several teams, test the problem's tags in each task's custom condition. `entity_tags` is an array, so these are exact matches (§2).

```
tasks:
  notify_checkout:
    name: notify_checkout
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ "owner:checkout" in (event().get("entity_tags") or []) or
"owner:cart" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#checkout-alerts"
      message: "{{ event()['event.name'] }}"

  notify_payments:
    name: notify_payments
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ "owner:payments" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#payments-alerts"
      message: "{{ event()['event.name'] }}"

  notify_platform:
    name: notify_platform
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ "owner:platform" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#platform-alerts"
      message: "{{ event()['event.name'] }}"
```

> **Sources:** Assign team ownership (DT docs), Problem and event triggers (DT docs).

Management Zone Routing (legacy — do not build new workflows on this)

> ⚠️ **This pattern has nothing to match.** `management_zones` is **not present on problem records**: on a live tenant (10/05/2026) 0 of 15,878 problems in `dt.davis.problems` over 30 days carried it, while 14 of 15 hosts belonged to a management zone. A condition that reads it has nothing to evaluate against, whether or not your zones still exist. There is no Management Zone filter on the problem trigger itself (WFLOW-02 §2).
>
> Use the ownership-tag pattern above instead. If a region is the routing dimension, tag the entities (`region:us-east`) or use Smartscape ownership rather than reading the MZ array. Teams actively migrating off MZs should read MZ2POL-01 §5.

To confirm on your own tenant: `fetch dt.davis.problems, from:-30d | summarize n = countIf(isNotNull(management_zones))` .

Dynamic Channel Selection

Use JavaScript to determine the channel dynamically:

```
import { execution } from '@dynatrace-sdk/automation-utils';

export default async function () {
  const ev = (await execution()).params.event;   // trigger payload
  // entity_tags still works but is deprecated (Dynatrace Classic) - see the
upgrade guide
  const tags = ev.entity_tags || [];            // "key:value" strings

  // Find the ownership tag
  const ownerTag = tags.find(t =&gt; t.startsWith('owner:'));
  const team = ownerTag ? ownerTag.split(':')[1] : 'platform';

  return {
    channel: `#${team}-alerts`,
    team: team
  };
}
```

> **Sources:** Upgrade guide — alert notification (DT docs) — *"entity_tags still exists but is deprecated (Dynatrace Classic)"*.

5. Time-Based Routing

Business Hours vs Off-Hours

`now()` is UTC unless you pass a time zone: *"If no timezone is provided, UTC is used."* Without one, 09:00–17:00 below would mean 09:00–17:00 UTC, which is 05:00–13:00 in New York, and the weekday boundary would fall at midnight UTC. Pass the team's zone (a pytz name) to every time check:

```
tasks:
  # Business hours (New York): Slack channel
  slack_business_hours:
    name: slack_business_hours
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ now("America/New_York").weekday() &lt; 5 and
now("America/New_York").hour &gt;= 9 and now("America/New_York").hour &lt; 17
}}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-production"
      message: "{{ event()['event.name'] }}"

  # Off-hours: PagerDuty for Critical only
  pagerduty_off_hours:
    name: pagerduty_off_hours
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (now("America/New_York").weekday() &gt;= 5 or
now("America/New_York").hour &lt; 9 or now("America/New_York").hour &gt;= 17)
and (event().get("event.severity") | int(5)) &lt;= 1 }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"
      eventAction: trigger
      severity: critical
      summary: "[Off-Hours] {{ event()['event.name'] }}"
      source: dynatrace
      dedupKey: "dynatrace-{{ event()['display_id'] }}"

  # Off-hours, not Critical: queue for morning
  slack_queue:
    name: slack_queue
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (now("America/New_York").weekday() &gt;= 5 or
now("America/New_York").hour &lt; 9 or now("America/New_York").hour &gt;= 17)
```

```
and not ((event().get("event.severity") | int(5)) &lt;= 1) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-queue"
      message: ":moon: *Queued for review* {{ event()['event.name'] }}"
```

For holidays and irregular hours, the same reference documents business calendars, read with `calendars()`.

Weekend-Only Routing

```
tasks:
  weekend_oncall:
    name: weekend_oncall
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ now("America/New_York").weekday() &gt;= 5 and
(event().get("event.severity") | int(5)) &lt;= 1 }}'
      else: SKIP
    input:
      # Page the weekend on-call rotation: a connection that holds that
rotation's
      # routing key. Credentials live in connections, not in expressions.
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-weekend-oncall') }}"
      # ... event inputs as in §3
```

> **Sources:** Jinja expressions for Workflows (DT docs) — *"The current timestamp in UTC or time zone is provided as the input parameter."*

6. Escalation Patterns

Timed Escalation

Escalate if the problem is still active after a time window. This pattern checks the Dynatrace problem's `event.status`; it does not see whether anyone acknowledged the

alert in Slack or PagerDuty. In community practice, acknowledgment-aware escalation is left to the paging tool's own escalation policy.

```
tasks:
  # Step 1: Initial notification
  initial_slack:
    name: initial_slack
    action: dynatrace.slack:slack-send-message
    predecessors: []
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-production"
      message: ":warning: {{ event()['event.name'] }} - paging on-call if
still active in 15 min"

  # Step 2: After 15 minutes, check if still open (event.status is ACTIVE or
  # CLOSED — never OPEN). There is no "wait" action: the delay is this task's
  # own Wait before option, in seconds (max 86400).
  check_problem_status:
    name: check_problem_status
    action: dynatrace.automations:run-javascript
    predecessors:
      - initial_slack
    waitBefore: 900
    input:
      script: |
        import { execution } from '@dynatrace-sdk/automation-utils';
        import { queryExecutionClient } from '@dynatrace-sdk/client-query';

        export default async function () {
          const ev = (await execution()).params.event;   // snapshot from
trigger time
          // Re-query: the trigger payload is 15 minutes old by now
          const started = await queryExecutionClient.queryExecute({ body: {
requestTimeoutMilliseconds: 30000, query:
            `fetch dt.davis.problems, from:-2h | filter display_id ==
"${ev['display_id']}" | sort timestamp desc | limit 1 | fields event.status` }
});
          let q = started;   // no result yet if the query outlasted the
timeout - poll (see WFLOW-08 §2)
          while (q &amp;amp;&amp;amp; (q.state === 'NOT_STARTED' || q.state ===
'RUNNING')) {
            q = await queryExecutionClient.queryPoll({ requestToken:
started.requestToken, requestTimeoutMilliseconds: 30000 });
          }
          if (!q || q.state !== 'SUCCEEDED') throw new Error(`DQL query did
not succeed: ${q?.state ?? 'no response'}`);
          return { escalate: q.result.records[0]?.['event.status'] ===
'ACTIVE' };
```

```
        }

  # Step 3: Escalate to PagerDuty
  escalate_pagerduty:
    name: escalate_pagerduty
    action: dynatrace.pagerduty:send-event
    predecessors:
      - check_problem_status
    conditions:
      states:
        check_problem_status: OK
      custom: '{{ result("check_problem_status").escalate }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"
      eventAction: trigger
      severity: error                 # Events API v2 accepts critical,
error, warning, info
      source: dynatrace
      dedupKey: "dynatrace-{{ event()['display_id'] }}"
      summary: "[ESCALATED] {{ event()['event.name'] }} - still active after
15 min"
```

Workflows has no separate wait action. **Wait before** is a task option: *"The Wait before option controls how long a task stays in the waiting state before being run."* It takes seconds, up to 86,400, or a Jinja expression.

The same effect needs no waiting task at all: create a second workflow whose Problem trigger has **Minimum duration** 15 minutes and holds only the PagerDuty task. *"Minimum duration : Postpones the trigger until the problem has been open for at least the configured duration."* A problem that closes within 15 minutes never starts it.

> **Sources:** Build workflows — Task options (DT docs) — *"The default Wait is 0 second and max value is 86400 seconds."*; PagerDuty Connector actions (DT docs) — *"Trigger, acknowledge, or resolve an alert in PagerDuty using the Events API v2"*; Problem and event triggers (DT docs).

Multi-Tier Escalation

```
0 min  → Slack channel notification
15 min → Email to team lead
30 min → PagerDuty to on-call
60 min → PagerDuty to manager
```

7. Multi-Channel Strategy

Recommended Channel Matrix

`event.severity`	Environment	Channel(s)
1 (Critical)	Production	PagerDuty + Slack urgent + Email
1 (Critical)	Staging	Slack alerts
2 (Major)	Production	Slack alerts + Email
2 (Major)	Staging	Slack alerts
3 (Minor)	Production	Slack alerts
3 (Minor)	Staging	Slack (daily digest)
unset (SaaS 1.348+)	Any	Slack (daily digest), until the source sets a severity

Levels 4 (Warning) and 5 (Informational) do not raise problems on their own (§3), so a problem-triggered workflow needs no row for them.

Complete Multi-Channel Workflow

```
tasks:
  # PagerDuty: Critical + Production
  pagerduty:
    name: pagerduty
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 and
"env:prod" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"
      # ... event inputs as in §3

  # Slack Urgent: Critical + Production
  slack_urgent:
    name: slack_urgent
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 and
"env:prod" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-urgent"
      message: ":rotating_light: {{ event()['event.name'] }}"

  # Slack Standard: Major or above + Production
  slack_standard:
    name: slack_standard
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 2 and
"env:prod" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
```

```
      channel: "#alerts-production"
      message: "{{ event()['event.name'] }}"

  # Email: Critical, or Major + Production
  email_alert:
    name: email_alert
    action: dynatrace.email:send-email
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 or
((event().get("event.severity") | int(5)) == 2 and "env:prod" in
(event().get("entity_tags") or [])) }}'
      else: SKIP
    # input: to, subject, content
```

Query Workflow Routing Effectiveness

```
// Workflow executions by outcome
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`,
and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in
any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to
`dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by
`event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION ·
WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript,
send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR ·
DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it
is ERROR
//   .state.is_final                    true only on terminal records —
filter on it, otherwise a
//                                        single execution is counted once as
RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                        (was task.error)  ·  duration (was
task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT"
| limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "WORKFLOW_EXECUTION"
| filter dt.automation_engine.state.is_final == true
| summarize executions = count(), by:{dt.automation_engine.state}
| sort executions desc
```

```
// Task execution distribution by notification channel
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`, and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to `dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by `event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION · WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript, send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR · DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it is ERROR
//   .state.is_final                     true only on terminal records — filter on it, otherwise a
//                                        single execution is counted once as RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                         (was task.error)  ·  duration (was task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT" | limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "ACTION_EXECUTION"
| filter dt.automation_engine.state.is_final == true
| summarize executions = count(), by:{dt.automation_engine.action.app, dt.automation_engine.action.function}
| sort executions desc
| limit 20
```

```
// Skipped tasks (conditions not met)
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`, and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to `dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by `event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION · WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript, send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR · DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it is ERROR
//   .state.is_final                     true only on terminal records — filter on it, otherwise a
//                                        single execution is counted once as RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                         (was task.error)  ·  duration (was task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT" | limit 1
fetch dt.system.events, from:-24h
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "TASK_EXECUTION"
| filter in(dt.automation_engine.state, {"SKIPPED", "DISCARDED"})
| summarize skipped = count(), by:{dt.automation_engine.workflow.title, dt.automation_engine.task.name, dt.automation_engine.state}
| sort skipped desc
| limit 25
```

Next Steps

With routing configured, integrate with incident management:

Recommended Path

1. **WFLOW-05: PagerDuty & ServiceNow** - Create incidents automatically
2. **WFLOW-06: Custom Templates** - Rich message formatting
3. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation

Key Takeaways

- **Conditions** live on each task: predecessor states, one custom Jinja expression, and an Else of Stop or Skip
- **Severity routing** keys on `event.severity` 1 Critical / 2 Major / 3 Minor; problems never carry 4 or 5
- **Team routing** uses `owner` / `dt.owner` tags (best in the trigger's tag filter) or Smartscape ownership — not management zones
- **Time-based routing** reduces off-hours noise; pass a time zone to `now()`, which is UTC otherwise
- **Escalation patterns** prevent missed alerts
- Test conditions with On-Demand trigger before production

Summary

In this notebook, you learned:

- Why and when to route alerts conditionally
- Conditional expression syntax and operators
- Severity-based routing patterns
- Team routing by entity tag, and why management-zone routing is now legacy
- Time-based (business hours) routing
- Escalation with wait-and-check patterns, or a second workflow with a Minimum duration
- Multi-channel notification strategy

References

- Workflow reference / Jinja expressions + conditions (DT docs)
- Build workflows — task conditions and options (DT docs)
- Problem and event triggers (DT docs)
- Standardized event severity (DT docs)
- Workflow actions umbrella (DT docs)
- Notification actions umbrella (DT docs)
- Davis Problems app (DT docs)
- Alerting and notifications umbrella (DT docs)
- Upgrade guide — alerting and notifications (DT docs)

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